

Rank Before You Merge: A Stage Law for Training-Free Visual Token Compression in Merger-Equipped Vision-Language Models

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Abstract

Native spatial mergers reduce visual-token count before the language model, but their interaction with token selection remains under-controlled. We compare query-blind L2 ranking before and after the merger at matched budgets. The operational RBM implementation (Qwen3-VL layer 8) leads matched Post-L2 on all nine TextVQA, DocVQA, and OCRBench cells across three models. A separate full-split Qwen3-VL control at the main-merger boundary reveals a workload-conditioned stage effect: pre-final leads post by +27.7 pp on TextVQA, +4.6 pp on DocVQA, and +235 OCRBench points, whereas post leads by 5.6 pp on GQA; all four paired contrasts remain significant after Holm correction.

On Qwen3-VL, the merger weakly preserves unit ranks and disproportionately demotes edge-rich document units; swapping the pre ranking into the post path recovers pre accuracy with identical selected-unit identities. We turn this into **Rank-Before-Merge (RBM)**, a minimal, training-free stage change with no learned scorer or query input. The gains are against the matched L2-family post arm, not a universal method ranking: query-conditioned FastV remains stronger on several reference-grounded and object-centric settings. On full OCRBench under the same HF eager harness and 4M-pixel cap, RBM scores 559 vs 413 on Qwen3-VL and 382 vs 295 on Qwen2.5-VL. RBM is therefore an OCR-oriented default rather than a universal winner. Six prespecified extensions do not improve over their stronger constituents. At 25% retention, compression is stage-neutral in speed, increasing throughput by 68% on Qwen3-VL and 1.8–2.5 $\times$ on InternVL3.

Keywords

visual token compression, vision-language models, training-free inference, multimodal foundation models, inference efficiency

1 Introduction

Multimodal inference cost is a visual-token problem. Visual-language models (VLMs) process images as hundreds to tens of thousands of visual tokens, making quadratic attention and key-value memory major inference costs. Visual-token compression is therefore central to efficient document understanding, scene-text QA, and OCR. In merger-equipped architectures such as Qwen-VL and InternVL3, 2×2 patch groups are first merged, while prior work focuses mainly on which tokens to keep and how to score them. The selection stage, before or after the native merger, is rarely isolated as a matched variable. Prior work notes Qwen’s native token reduction [3] and OCR degradation from post-encoder merging [13, 41]; we directly test the boundary by scoring the same units with query-blind L2 on merger-input (pre) or merger-output (post) features at the same per-image budget.

Minimal operationalization: Rank-Before-Merge (RBM). Score each merger-input unit, retain the top- κ , then merge only the survivors. RBM uses no attention weights, training, or query

input. Because merging is per-unit, a fixed kept set receives the same native per-unit merger operation; byte identity is verified on Qwen3-VL. RBM changes which units reach the merger. We use this rule to expose the stage variable, not to claim scorer novelty (§3.3).

What falls out of it — a workload-conditioned stage law.

The operational layer-8 RBM arm leads matched Post-L2 on the evaluated text-dense splits across two Qwen generations; the direction replicates on InternVL3. Separately, the full-split matched-boundary Qwen3-VL control significantly favors pre-merger ranking on TextVQA, DocVQA, and OCRBench, but significantly favors post-merger ranking on GQA (§4).

Why the stage matters. On Qwen3-VL, the learned merger weakly preserves unit ranks (Spearman $\rho = 0.14\text{--}0.36$), disproportionately demotes high-edge units on documents, and yields to a ranking-swap control: the post path recovers pre accuracy with identical selected sets (Jaccard = 1.000 in all four tested cells). Thus, where tested byte-exactly, the ranking rather than the forward path explains the gap (§5). This matched-rule result is distinct from the query-conditioned FastV comparison: FastV remains stronger on several scene-text/object-centric cells, so RBM is a query-blind, OCR-preserving robust default, not a universal winner (§4.4).

Contributions. First, Qwen3-VL diagnostics connect merger-induced rank reshaping and edge-rich unit demotion to the selected set (§5). Second, a full-split matched-boundary control establishes a workload-conditioned stage effect, with opposite preferences for text-centric tasks and GQA (§4). Third, we evaluate minimal RBM across three models, map its trade-off against FastV, and report six unsuccessful extensions (§4.4, §6).

2 Related Work

Token pruning and merging. VLM compressors select or merge visual tokens at encoder outputs or inside early LLM layers. FastV uses LLM attention and is our query-conditioned baseline [6]; PyramidDrop prunes progressively [39]. FasterVLM and SparseVLM accelerate or condition pruning [48, 49]; PruMerge and FitPrune use merging or learned importance [29, 42]. Other selectors use visual attention, graph propagation, diversity, or text-guided cues [1, 10, 14, 25, 34, 43]. These approaches motivate strong baselines but do not match selection on the two sides of a frozen model’s native merger under a shared query-blind scorer.

Merger-aware compression. ToMe and AdaptMerge introduce an additional token-merging operation [4, 13]; learned projectors such as TokenPacker alter the visual interface [21]. VisionZip’s inspected Qwen2.5-VL path selects after PatchMerger [41]; AnchorPrune likewise operates after Qwen2.5-VL’s model-specific spatial processing [28]. GlimpsePrune, VScan, TransPrune, and V2Drop provide complementary Qwen-family or layer-evolution comparisons [5, 19, 44, 45], but not the matched native-merger boundary control studied here.

Kept visual units (blue) per selector — two OCR cases

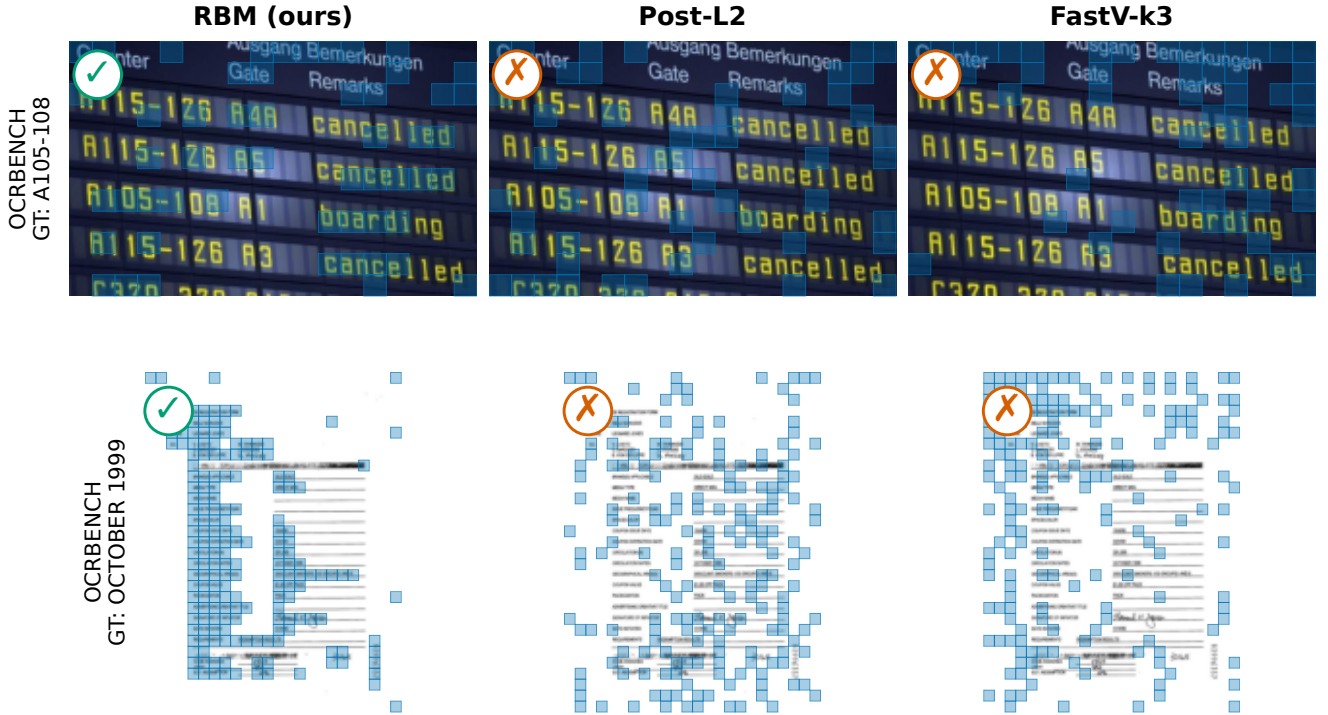


Figure 1: Three selectors retain different visual evidence. Two audited OCRBench cases compare RBM, Post-L2, and FastV-k3 at the same $\kappa = 25\%$ visual-token budget on Qwen3-VL-8B. Blue native-grid cells are the visual units retained by each method; the underlying input remains visible for direct spatial comparison. Vector checks and crosses report answer correctness against the printed ground truth. In both examples, RBM preserves the answer-bearing text and answers correctly, while Post-L2 and FastV-k3 do not. These cases illustrate the selection behavior; aggregate effects and uncertainty are reported in Figure 2.

Recent design space. Hierarchical selection and set coverage broaden ranking [26, 33, 40]. Reconstruction, functional sensitivity, and learned conditioning offer other objectives [9, 15, 17, 31, 32, 37, 51]. Other work varies token lifetime or aggregates discarded content [7, 18, 20, 22, 36, 46]; document-specific studies emphasize that aggregate accuracy can conceal lost textual evidence [8, 23]. QuietPrune acts inside the ViT [11]; PixelPrune acts before it [35]. FastV and PyramidDrop act later inside the LLM. Our narrower question is where to rank native merge units while leaving the model and merger unchanged.

Evaluation. We use official scorer implementations from the lmms-eval family [47] and confine throughput comparisons to one engine, consistent with protocol-sensitive efficiency evaluation [38].

3 Method — Rank-Before-Merge (RBM)

3.1 Models, mergers, and the stage axis

We study three merger-equipped VLMs with two merger designs. Qwen3-VL-8B and Qwen2.5-VL-7B use a learned PatchMerger [2, 3]; InternVL3-8B uses pixel-shuffle downsampling [50]. All are served

in bf16, eager attention, vLLM 0.19 (V1) [16], single A40 (46 GB), greedy decoding. The shared pipeline is: image $\rightarrow$ ViT patch tokens $\rightarrow$ native merger (2×2 groups $\rightarrow$ one token) $\rightarrow$ language model. The Qwen merger is *not* an averaging pool: the four patch features of each 2×2 group are concatenated and projected by a small learned MLP (LayerNorm $\rightarrow$ fc1 $\rightarrow$ GELU $\rightarrow$ fc2) to one unit vector; the lossy step is the four-to-one nonlinear projection, so a kept unit’s representation is a nonlinear function of its four patches. We verified this against the served implementation on both Qwen generations.

The stage axis (the experimental variable). The two arms share model, token count, decoding, and a query-blind L2 magnitude rule, while evaluating that rule on the native feature state available on each side of the merger:

- **Pre-merger (RBM).** For each 2×2 unit, average the L2 norms of its four merger-input patch vectors, keep the top- κN , and pass only the survivors through the unmodified native merger (and, for Qwen3-VL, each deepstack merger). The merger operates exactly as in the uncompressed model, on a subset of units.

- **Post stage (contrast).** Either score the N merged units on merger-*output* features and keep the top- κN (the stage used by published compression methods for merger-equipped VLMs, including VisionZip’s Qwen path), or — the stronger, query-conditioned contrast — prune inside the LLM by layer-2 attention (the FastV stage). The main matrix (Table 1) uses the post-merger-L2 contrast from the same query-blind L2 family; the layer-2 FastV contrast is used only in Table 3.

Budget definition. Retention κ is defined over *merge units* relative to each image’s **own** full unit count, so two methods at the same κ feed the language model the same number of visual tokens per image; we verify equality of the mean post-merger token count per benchmark (**iso-token** control), so any accuracy difference is attributable to *which* units survive and the representational state on which selection is made, not to token count. We report $\kappa \in \{0.25, 0.125\}$ (25% / 12.5% retention); per-benchmark token counts are printed with each table.

3.2 The L2 selector as a control variable

We deliberately use the simplest **text-agnostic, query-blind** L2-magnitude family. For unit u with four merger-input patch vectors $h_{u,1:4}$ and native merger M , the implemented scores are

$$s_{\text{pre}}(u) = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2, \quad s_{\text{post}}(u) = \|M(h_{u,1:4})\|_2.$$

Thus the ranking functional is matched at the level of feature magnitude, while its representation is the experimental stage: four input vectors before the nonlinear merger versus its output vector after it. For Qwen3-VL, the headline operational pre arm takes h from the first deepstack tap (ViT layer 8), whereas the matched-depth pre-final control takes h from the final ViT block at the main-merger boundary (§4.2). We freeze plain RBM and add no deployed variant. §5 additionally reports a mechanism-side probe: a *variance-weighted* variant of the same query-blind scorer, $s_{\text{freq}}(u) = \alpha z(\|f_u\|_2) + \beta z(\text{Var}_u)$ (both terms per-image standardized, so the ratio $\alpha:\beta$ is a genuine trade-off knob rather than an arbitrary scale), improves text-dense accuracy under aggressive compression (+1.2 pp TextVQA at 25% retention, +0.5 pp at 50%) yet is not a general upgrade: it regresses −1.6 to −2.2 pp on DocVQA, OCR-Bench, and GQA at 25%. We therefore treat it as supporting evidence for §5.2’s high-frequency-unit hypothesis rather than as a deployed selector variant (§6). §5.4 reports that the tested effect survives replacing L2 with a second scorer family (global-centroid attention) on Qwen3-VL, indicating the effect is not specific to L2. Unlike the merging lineage of §2, RBM introduces no second learned merger and no query input: it manipulates only the *set* of units the native merger sees — which is exactly what makes the pre-vs-post contrast a clean stage experiment rather than a scorer comparison.

3.3 Rank-Before-Merge

Given an image with N merge units, compute $s(u)$ on merger-input features for every unit, retain the top $k = \kappa N$, and invoke the native merger(s) on the retained units only. Unit identity is shared across the main and deepstack streams (deepstack mergers process intermediate-depth features of the *same* units), so pruning a unit

Algorithm 1 Rank-Before-Merge (RBM), one image

Require: image x ; retention κ ; query-blind scorer $s = \|\cdot\|_2$

Ensure: compressed visual token sequence z

```

1:  $P \leftarrow$  ViT patch tokens of  $x$  {16 px footprint}
2:  $U \leftarrow$  group  $P$  into  $N$  native  $2 \times 2$  units {32 px footprint}
3: for each unit  $u \in U$  do
4:    $\text{score}[u] \leftarrow \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2$  {mean patch L2}
5: end for
6:  $k \leftarrow \max(1, \text{round}(\kappa N))$ ;  $K \leftarrow \text{topk}(\text{score}, k)$ 
7:  $z_{\text{main}} \leftarrow \text{NativeMerger}(\{u : u \in K\})$  {merge survivors only}
8:  $z_{\text{ds}} \leftarrow \text{DeepstackMergers}(\{u : u \in K\})$  {Qwen3-VL only}
9:  $z \leftarrow \text{NativeVisualInterface}(z_{\text{main}}, z_{\text{ds}})$ 
10: return  $z$  {feed standard LLM}
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removes it from every stream. No attention weights are required and no parameters are trained.

Unit equivalence (the structural invariant). Because the mask is at *unit* granularity and the merger is per-unit, a fixed kept set supplies the same four patches to the same native per-unit operation. Selection cannot prevent intra-unit combination; it changes which units reach the merger. The resulting outputs are byte-identical in the verified Qwen3-VL path; other families are limited to the evidence stated in §8. Pre-merger selection keeps units on their raw (merger-input) saliency; post-stage selection keeps them on a saliency the merger has already rewritten (§5). This invariant enables the ranking-swap control of §5.3.

Post-merger selection (the contrast): replace lines 3–5 of Algorithm 1 by $U' \leftarrow \text{NativeMerger}(U)$; $\text{score}'[u'] \leftarrow \|f_{\text{out}}(u')\|_2$ on merger-*output* features; $K' \leftarrow \text{top-}k$ by score' ; keep(K'). Everything downstream is identical.

Configuration details. Per-family tap points (Qwen3-VL deepstack[0] at ViT layer 8, Qwen2.5-VL main merger input, InternVL3 pixel-shuffle input) and the Qwen2.5-VL position-cursor fix are detailed in Supplementary § S8, including the cross-generation configuration diagnostic.

4 Main Experiments

4.1 Setup

Benchmarks and metrics. We use TextVQA VQA accuracy [30], DocVQA ANLS [27], the official OCR-Bench score [24], and GQA normalized exact match [12]. Official split sizes are 5000, 5349, 1000, and 12578, respectively. Scorers are verbatim ports of the lmms-eval family [47], with ground-truth self-test passing 200/200.

Engines and fairness protocol. RBM and the none / post cells run in vLLM; the FastV baseline runs in an independent HuggingFace transformers eager harness (it has no vLLM path). Accuracy is comparable across engines (the harness bit-degrades to native at zero drop, 8/8 per-sample identical; HF-vs-vLLM none cells agree 16/16); **throughput is not** — all efficiency numbers are confined to vLLM (§7). Fairness is enforced by the same keep ratio relative to each image’s own token count and by reporting mean post-merger token counts.

Significance. Each pre-vs-post Δ pairs the same samples under greedy decoding. The prespecified primary family comprises the nine model×text-dense-benchmark pre-vs-post contrasts at

25% retention in Table 1. We use 20,000-resample paired-bootstrap confidence intervals on per-item metric differences and two-sided paired sign-flip permutation tests; Holm correction across these nine tests gives $p_{\text{Holm}} \leq 4.5 \times 10^{-4}$ for every contrast. McNemar is reported only as a binary-outcome cross-check for OCRBench containment and GQA exact match. Other budgets, GQA, mechanism probes, FastV comparisons, and extension gates are secondary or exploratory and are not pooled into the primary family. The metric-level tests preserve partial credit in VQA accuracy and ANLS rather than binarizing those metrics. The four full-split matched-boundary contrasts form a separate prespecified family; all remain significant after Holm correction within that family.

4.2 Main result across three models — official full splits (Table 1)

Operational results. Every text-dense Δ_{25} is large and significant: RBM leads post-merger selection by +38.4 pp (TextVQA), +24.3 pp (DocVQA), +363 pts (OCR-Bench; 547 vs 184, 3.0 $\times$) on Qwen3-VL, and by +26.1 pp, +11.0 pp, and +298 points (480 vs 182) on Qwen2.5-VL; all paired-bootstrap 95% CIs exclude zero. The matched-cap Qwen2.5-VL OCRBench values are 480 vs 182 (+298 points) at 25% and 328 vs 70 (+258 points) at 12.5%. The gap is robust to deeper compression: at $\kappa = 0.125$ the post stage degrades much faster (Qwen3-VL OCR-Bench 184 $\rightarrow$ 53 vs RBM 547 $\rightarrow$ 350, widening the pre/post ratio from 3.0 $\times$ to 6.6 $\times$), and all six text-dense $\Delta_{12.5}$ remain significant under paired metric-level inference.

Full-split matched-boundary control. Operational Qwen3-VL RBM ranks at the layer-8 deepstack[0] input. To isolate the main-merger boundary, pre-final instead ranks final ViT-block features immediately before that merger, while post ranks its output. Both arms use $\kappa = 0.25$, identical per-sample token counts, greedy decoding, and zero skips.

The control reveals a workload-conditioned stage effect: text-, document-, and OCR-centric tasks favor pre-merger ranking, whereas GQA significantly favors post-merger ranking. Operational layer-8 RBM improves GQA over pre-final (.4488 vs .4207) but remains below post (.4771), so neither operational nor matched-boundary evidence supports a universal pre-merger advantage. The no-merger LLaVA-1.5 control is summarized in Supplementary S7c.

4.3 Replication on a third family — InternVL3-8B (Table 1)

To test whether the Qwen direction replicates beyond that family, we run RBM and the post stage on **InternVL3-8B**, whose merger is pixel-shuffle downsampling — a different design from Qwen’s PatchMerger. The post arm here is **post-merger L2 top-k** — the same query-blind L2-magnitude family as the pre arm, at an identical token budget (ptid equal across arms), so the InternVL3 block of Table 1 isolates the before/after merger stage under iso-budget control at 25% retention. The query-conditioned in-layer FastV contrast is a separate campaign, reported in Table 3.

Configuration boundary. InternVL3 DocVQA uses a 4 M-pixel cap, while the Qwen same-scope baseline cells of §4.4 use 600k; cross-model absolute values are not comparable. At 25% retention RBM costs −19.4 pp vs none on InternVL3 DocVQA and −99/1000 on OCR-Bench, versus only −4.5 pp on TextVQA and −3.0 pp on GQA.

The claim is therefore the large within-model pre>post margin, not losslessness on every text-dense workload.

At 12.5% retention ($r = 0.875$), TextVQA is 0.7230 pre vs post 0.3064 (+**41.7 pp**); DocVQA pre 0.5054 vs post 0.2451 (+**26.0 pp**) — the gap *widens* under deeper compression, as on Qwen.

Cross-family directional replication. InternVL3’s pre–post text-dense gaps (+37.4/+34.6 pp / +432 pts) are the same order and direction as Qwen3-VL’s (+38.4/+24.3/+363 pts) on all three text-dense benchmarks, and GQA remains near zero and not significant here (−0.4 pp), whereas Qwen3-VL’s matched-boundary control significantly favors post by 5.6 pp (§4.2), with no text-dense crossover at either family’s native imaging configuration. These accuracy-level replications show that the shared L2-magnitude rule can favor the pre-merger side on a third model family with a different merger. They support — but do not by themselves establish — the Qwen3-VL selection-level mechanism as a cross-family explanation.

The ranking-swap comparison on InternVL3. The M3-style control (§5) — run the post-merger forward path but select units with the pre-merger ranking — closely matches pre-merger accuracy on the three tested InternVL3 cells: swap vs pre is +0.00 pp (TextVQA), +0.11 pp (DocVQA), and −0.50 pp (GQA), with kept-set Jaccard(pre,swap)= 1.0 and per-sample answer-identity 0.98–1.0 ($n = 200$, iso-token, no swap fallback). This supports a ranking-level explanation on a second merger architecture (pixel-shuffle, not just Qwen’s PatchMerger), but the nonzero residuals mean that the byte-exact causal attribution remains specific to Qwen3-VL.

4.4 A workload-conditioned regime map — RBM vs the query-conditioned strong baseline FastV (Table 3)

FastV is the query-conditioned strong baseline in this comparison: it selects inside the LLM after cross-attention has mixed in the question. We compare RBM against the best tested FastV layer ($K = 3$), with paired samples and official rescoring within each row. We read the result as a **workload-conditioned regime map** — where query-conditioned selection wins, where it fails, and why — not as a single-winner race. At its canonical progressive schedule PyramidDrop is near-lossless but occupies a different budget point (equivalent mean retention 0.625 = 2.5 $\times$ ours); at the tested iso-25% schedule it degrades sharply (TextVQA 0.073 / GQA 0.305 / OCR 0.005 on Qwen3-VL), whereas RBM holds (0.60 / 0.42 / 0.58). There is therefore no universal winner.

Workload-conditioned comparison. FastV-k3 leads RBM on the three Qwen3-VL non-OCR rows (+17.2/+8.9/+16.2 pp) and Qwen2.5-VL TextVQA (+7.8 pp), showing the value of question-conditioned selection. Conversely, on full OCRBench RBM leads by 146/87 points on the /1000 scale; paired inference uses only the 921/912 samples answered by both methods. The Qwen2.5-VL $n = 200$ GQA/DocVQA differences remain inconclusive. Thus later query conditioning does not recover RBM’s OCR accuracy under the tested control, while it is beneficial on several non-OCR workloads.

Table 1: Operational RBM versus matched Post-L2 on official full splits ($n = 5000/5349/1000/12578$). $\Delta_{25} = \text{RBM} - \text{post}$ at $\kappa = 0.25$. Qwen3-VL RBM ranks at the layer-8 deepstack tap; this table is therefore an operational comparison, distinct from the matched-boundary control in Table 2. Within each row the two arms use equal token budgets; full protocol and skip accounting are in the supplementary material.

Model	Benchmark (metric)	none	RBM @25%	post @25%	Δ_{25}	RBM @12.5%	post @12.5%
Qwen3-VL-8B	TextVQA (VQA-acc)	0.844	0.605	0.222	+38.4 pp	0.472	0.132
	DocVQA (ANLS)	0.956	0.481	0.238	+24.3 pp	0.352	0.103
	OCR-Bench (/1000)	760	547	184	+363 pts	350	53
	GQA (exact match)	0.616	0.449	0.477	$-2.8 \text{ pp } \uparrow$	—	—
Qwen2.5-VL-7B	TextVQA (VQA-acc)	0.862	0.702	0.442	+26.1 pp	0.597	0.319
	DocVQA (ANLS)	0.949	0.636	0.526	+11.0 pp	0.455	0.245
	OCR-Bench (/1000)	817	480	182	+298 pts	328	70
	GQA (exact match)	0.604	0.559	0.585	$-2.6 \text{ pp } \uparrow$	—	—
InternVL3-8B	TextVQA (VQA-acc)	0.8338	0.7890	0.4148	+37.4 pp	0.7230	0.3064
	DocVQA (ANLS)	0.9221	0.7284	0.3820	+34.6 pp	0.5054	0.2451
	OCR-Bench (/1000)	852	753	321	+432 pts	—	—
	GQA (exact match)	0.6293	0.5993	0.6031	-0.4 pp (n.s.)	—	—

Notes. Qwen2.5-VL OCRBench uses the same 4M-pixel cap in both arms; mean effective prompt lengths are equal at each budget (229.6 at 25%, 131.5 at 12.5%). InternVL3 uses pixel-shuffle and family-specific imaging; compare within-model deltas. Detailed configuration, inference, and skips are in the supplementary material.

Table 2: Qwen3-VL full-split matched-boundary control. $\Delta = \text{pre-final} - \text{post}$; OCRBench differences and CIs are shown in /1000 points.

Benchmark	pre-final	post	Δ	95% CI	n
TextVQA	.4985	.2217	+27.68 pp	[+26.21, +29.19]	5000
DocVQA	.2836	.2377	+4.59 pp	[+3.43, +5.78]	5349
OCRBench	419	184	+235 pt	[+199, +271]	1000
GQA	.4207	.4771	-5.64 pp	[$-6.42, -4.86$]	12578

Equal per-sample token counts; zero skips in both arms. All four paired tests remain significant after within-family Holm correction; full statistics are in Supplementary S7b.

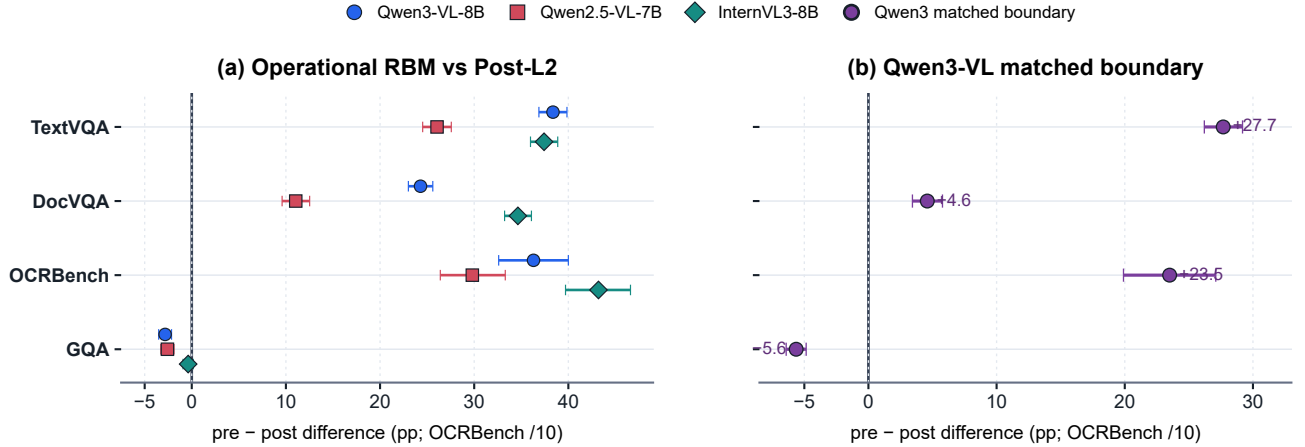


Figure 2: Operational and matched-boundary stage effects. (a) Operational RBM minus Post-L2 on full splits; Qwen3-VL uses its layer-8 RBM tap. (b) The separate Qwen3-VL control ranks immediately before versus after the main merger. Whiskers are paired-bootstrap 95% CIs; OCRBench /1000 differences are divided by 10. The matched control favors pre on text-centric workloads and post on GQA.

Table 3: Workload-conditioned comparison of FastV at best $K = 3$ (the query-conditioned strong baseline) vs RBM (pre, the query-blind OCR-oriented default) at 25% retention. Configuration is matched within each row; scope and pixel cap are printed explicitly.

Benchmark (scope; pixel cap)	n	FastV-k3	RBM @25%	Δ (F/R)
Qwen3-VL-8B				
TextVQA (full; default)	5000	0.7771	0.605	+17.2 F
GQA (full; default)	12578	0.5376	0.449	+8.9 F
DocVQA (dev; 600k cap)	200	0.5863	0.4239	+16.2 F
OCR-Bench (full; 4M cap)	1000	413	559	+146 R
Qwen2.5-VL-7B				
TextVQA (dev; default)	200	0.7467	0.6683	+7.8 F
GQA (dev; default)	200	0.475	0.520	+4.5 R[‡]
DocVQA (dev; 600k cap)	200	0.4852	0.5062	+2.1 R[‡]
OCR-Bench (full; 4M cap)	1000	295	382	+87 R

Notes. F = FastV; R = RBM. Only the OCRBench rows use the full 1000-example, 4M-cap, HF eager control: symmetric skips are 79 (Qwen3-VL) and 88 (Qwen2.5-VL), leaving paired answered sets of 921 and 912 for inference. OCRBench scores and deltas are reported on the /1000-point scale. Skips are identical across methods and cannot favor either arm. Other rows retain their printed full/dev scopes and configurations. [‡]The paired 95% CI crosses zero; full harness, budget, and skip details are in the supplementary material.

5 Mechanism — How the Native Merger Reshapes Saliency Rankings

We give the pre-vs-post gap a mechanistic account in three parts (M1–M3), measured on **Qwen3-VL-8B**. M1/M2 use deterministic seed-0 samples: OCRBench has $n = 199$, while TextVQA, DocVQA, and GQA each have $n = 64$; M3 uses $n = 200$ with the same invocation as the headline cells. We then state the architectural scope.

5.1 M1 — the merger reshuffles unit ranks

Per image we correlate the *pre* ranking (merger-input unit L2) with the *post* ranking (merged-token L2) over all units. The two rankings are barely related, and the reshuffle tracks text density: Spearman $\rho = 0.137$ (DocVQA, text-dense document) < 0.332 (TextVQA, text-dense scene) < 0.360 (GQA, object-centric), with the decision-relevant top-25% overlap rising in the same order (Jaccard@25% = $0.180 < 0.243 < 0.278$; full values in Supplementary S1). The merger’s nonlinear projection thus reorders unit saliency almost from scratch ($\rho = 0.14$ – 0.36 , nowhere near 1): both the full-ranking correlation and the decision-relevant top-25% overlap rise from the most text-dense benchmark to the object-centric one, so the reshuffle is largest on DocVQA and smallest on GQA — yet substantial on every benchmark.

5.2 M2 — demoted units are disproportionately edge-rich

With $\text{rank_shift} = \text{post_rank} - \text{pre_rank}$ ($+$ $\Rightarrow$ demoted) and per-unit Sobel edge energy as a high-frequency proxy, high-edge units are disproportionately demoted at $\kappa = 25\%$. The rank-shift/edge correlations are $+0.439$ on DocVQA, $+0.155$ on TextVQA, and $+0.036$ on GQA — $\sim 12\times$ stronger on documents than on object scenes. Units post drops that pre keeps are the highest-edge group: on DocVQA,

mean Sobel **0.641 vs 0.124** for the reverse group, with **92% above the per-image median (vs 35%)**. This is consistent with demotion of structures needed by text-centric recognition, but Sobel energy is not a text-region mask. The full M2 table (per-benchmark Sobel means and median fractions) is in Supplementary S2.

5.3 M3 — ranking-swap control with fixed token identities

By unit equivalence (§3.3), holding the *forward path* fixed at post and swapping in the *pre ranking* must reproduce pre accuracy if the gap is purely a ranking effect. It does:

Swapping in the pre ranking recovers full pre-merger accuracy (exactly on DocVQA; within 2/200 greedy-decode answers on TextVQA, ϵ -level kernel numerics confirmed by a 200/200 byte-identical independent rerun; full values are in Supplementary S9) and recovers the gap ($+38.3$ pp TextVQA, $+26.5$ pp DocVQA over post). The swap path agrees with post on only 40/200 (TextVQA) and 63/200 (DocVQA) answers — without the pre ranking, the same forward path produces the lower accuracy.

Kept-set identity on both architectures. Emitting the kept unit indices from both paths and comparing directly: **Jaccard(swap-kept, pre-top- κ) = 1.000 in all four cells** (Qwen3-VL and Qwen2.5-VL $\times$ DocVQA and TextVQA; $n = 32$; independent random 25%-retained sets have expected Jaccard ≈ 0.143). The two paths select *exactly the same units*. The small swap-vs-pre answer residual on Qwen2.5-VL ($+6.7$ pp DocVQA at $n = 32$) traces to an implementation confound of the swap control — window-attention reverse_indices (three recovery sites on Qwen2.5-VL, zero on Qwen3-VL) map identical index sets to physical units in different orderings — while batch-dependent merging is structurally excluded (the merger is per-token LN+MLP). On Qwen3-VL, fixing the selected token identities isolates the downstream effect of their ordering and forward-path placement (byte-identical); Qwen2.5-VL corroborates kept-set identity but retains an ordering-confound residual. The merged representations of kept units carry no stage-dependent loss on Qwen3-VL (byte-exact); on Qwen2.5-VL the kept set is identical but the ordering confound is not eliminated, so we scope the no-loss statement to the byte-exact family.

5.4 Failure modes and scope boundaries

DocVQA exhibits selection misdirection: post ranking moves away from text and the swap fully restores accuracy. TextVQA combines rank corruption with a budget effect, while GQA has near-null text directionality ($\rho = 0.04$). The stage effect is therefore workload-conditional. It retains sign and magnitude under a global-centroid attention scorer on Qwen3-VL, but that proxy degrades on Qwen2.5-VL; L2 remains the paper’s fixed selector. The ranking law replicates on Qwen2.5-VL and at the accuracy level on InternVL3 (Table 1), whereas the causal decomposition is claimed only for byte-exact Qwen3-VL. A low 600k pixel cap reverses shallow Qwen2.5-VL DocVQA; the native-resolution full split remains the headline configuration (§8).

Retention curve. The stages are nearly indistinguishable at 75% retention; their gap then widens monotonically through 25% on both Qwen models (full values in Supplementary S3, $n = 200$).

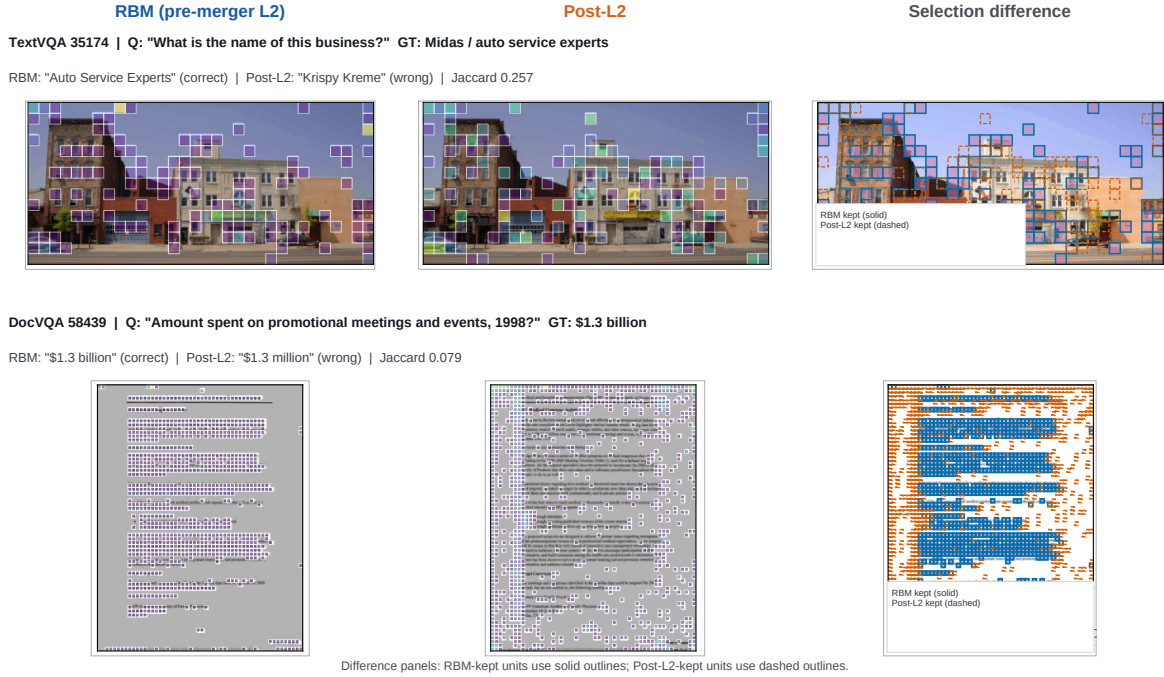


Figure 3: Measured token-survival maps at 25% retention. Rows show representative TextVQA and DocVQA examples; columns show units retained by operational RBM, Post-L2, and their selection difference. RBM preserves the correct answer in both cases, whereas Post-L2 does not. The maps reuse the recorded masks and predictions; they are qualitative examples, not aggregate accuracy evidence.

Merger mechanism. Figures 3 and 4 connect the stage effect to the native merger. Across four workloads, pre- and post-merger rankings share only moderate rank correlation and top-quartile overlap. Edge-rich, high-frequency units are disproportionately demoted after merging. Finally, a Qwen3-VL ranking-swap control keeps the selected unit set fixed (Jaccard = 1.000); fixing these identities isolates the downstream effect of ordering and forward-path placement.

6 Negative Results — Tested Extensions Do Not Generalize

Six prespecified Qwen3-VL extensions fail to improve consistently over their stronger constituent: image-level routing remains below always-pre; a merger-aware hybrid gains 2.3 pp on TextVQA but loses 8.0 pp on OCRBench; every nonzero query-embedding blend regresses; an RBM→FastV cascade misses the Pareto gate in all eight cells; a variance-weighted selector gains 1.2 pp on one TextVQA setting but regresses 1.6–2.2 pp elsewhere; and an adaptive stage router either reduces to always-pre or loses 7 pp on TextVQA. An exploratory learned boundary scorer similarly trades −16.0 pp TextVQA for +3.5 pp GQA. These results motivate the frozen L2 RBM variant and bound only the tested design space; complete grids, gates, and paired counts are in Supplementary S4b–S4c and S9.

7 Efficiency — Supporting Evidence

Efficiency is supporting evidence, measured only in vLLM 0.19 V1 offline on one A40. **Stage-neutral:** across 75/50/25% retention, pre vs post throughput is 4.61/4.52, 5.39/5.54, and 6.91/6.89 req/s; five-repeat measurements remain within 3% with no consistent direction. **Compression scaling:** relative to no compression, throughput rises 12/31/68% and 25%-retention latency falls from 0.36 to 0.23 s. InternVL3 independently yields 1.8–2.5× throughput at 25% retention, again with pre ≈ post at identical token counts. The flat ≈ 42 GB allocator reservation is not a memory measurement; continuous-batching efficiency remains future work.

8 Discussion and Limitations

The evidence supports RBM as an OCR-oriented default, not a universal winner: the matched-boundary control favors post on GQA, and FastV leads several question-conditioned cells. Four limitations delimit this conclusion. **(1) Architecture:** byte-exact M3 attribution is specific to Qwen3-VL; Qwen2.5-VL retains an ordering confound, InternVL3 provides bounded support, and models without a native merger are out of scope. **(2) Mechanism:** M1/M2 use a Sobel high-frequency proxy on $n = 64$ samples and M3 uses $n = 200$; no OCR boxes or text-region masks establish direct text semantics. **(3) Evaluation:** FastV uses HF eager while main RBM/Post-L2 cells use vLLM; full OCRBench controls share HF eager, but throughput remains vLLM-only. **(4) Configuration and inference:** cross-model absolute values are not comparable when imaging caps differ, and

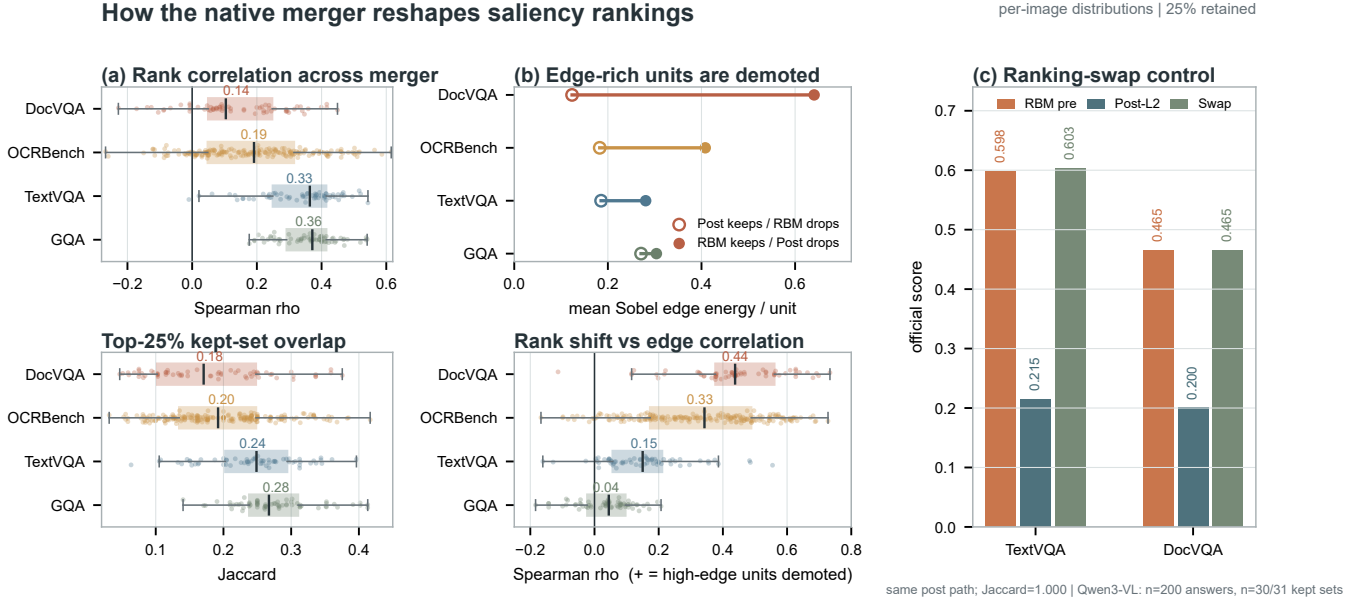


Figure 4: Quantitative evidence for saliency-rank reshaping. (a) Pre/post L2 rankings are only moderately preserved, measured by Spearman correlation and top-25% kept-set Jaccard across DocVQA ($n = 64$), OCRBench ($n = 199$), TextVQA ($n = 64$), and GQA ($n = 64$). (b) Rank shifts are associated with Sobel edge energy: units with stronger high-frequency structure are demoted more after merging, consistent with structures needed by text-centric recognition. (c) In the Qwen3-VL ranking-swap control, post-merger features use the pre-merger order while the kept set remains identical (Jaccard = 1.000); fixing token identities isolates the downstream effect of ordering and forward-path placement on $n = 200$ TextVQA and DocVQA samples.

secondary budgets, mechanism probes, baseline rows, and extension gates should be read within their stated families.

9 Conclusion

We separate the operational RBM implementation from a strict main-merger boundary control. Pre-merger ranking is consistently stronger on the evaluated text-, document-, and OCR-centric workloads, while full-split GQA significantly favors post-merger ranking. Qwen3-VL diagnostics support merger-induced saliency-rank reshaping as a selection-level mechanism. RBM is therefore a training-free OCR-oriented default rather than a universal winner: it exceeds FastV on full OCRBench, while FastV remains stronger on several question-conditioned settings. RBM adds no measured stage-specific serving cost, and six tested extensions do not consistently improve its trade-off.

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Supplementary Material

Rank Before You Merge: A Stage Law for Training-Free Visual Token Compression in Merger-Equipped Vision-Language Models

Every number is indexed by an audit identifier in S9. Tables S1–S3/S7 support § 5 of the body; S4–S6 support § 4 (S4b cascade counts support § 6); S8 supports § 3; S9 gives the per-cell artifact index, with cell, sample count, official metric, token count, skip accounting, and manifest identifier in S9.1–S9.4; S10 is a qualitative same-image pre/post selection comparison rendered from measured Qwen3-VL-8B merger features (four panels each, no paired question).

S1. M1 — merger ranking reshuffle, full table (Qwen3-VL-8B, seed-0, $n = 64$ /benchmark)

Per image, Spearman correlation of the pre ranking (merger-input unit L2) with the post ranking (merged-token L2) over all units, plus top-25% kept-set overlap. Evidence record: audit index entry for the M1 mechanism campaign. The Kendall- τ column is taken from the same audited M1 record.

Benchmark	Spearman ρ	Jaccard@25%	Kendall τ
DocVQA (document)	0.137	0.180	0.094
TextVQA (scene)	0.332	0.243	0.227
GQA (object)	0.360	0.278	0.249

Reading: the reshuffle is largest on the most text-dense benchmark and smallest (but still substantial) on object-centric GQA — $\rho = 0.137 < 0.332 < 0.360$; Jaccard = $0.180 < 0.243 < 0.278$. For two independent size- κN subsets, the large- N chance reference is $\kappa/(2 - \kappa) = 1/7 \approx 0.143$ at $\kappa = 25\%$. All three overlaps are above chance but low in absolute terms, with DocVQA closest to chance. Per-image distributions are in Supplementary Figure 5 (top).

S2. M2 — demotion directionality vs edge energy, full table (same sample as S1)

$\text{rank_shift} = \text{post_rank} - \text{pre_rank}$ ($+$ $\Rightarrow$ demoted); per-unit Sobel edge energy as a high-frequency proxy. Evidence record: anonymized artifact manifest entry for the M2 mechanism campaign. The per-benchmark Sobel means and median fractions are taken from the same audited M2 record.

Benchmark	ρ	Sobel drop	Sobel reverse	demoted > median
DocVQA	+0.439	0.641	0.124	92% (vs 35%)
TextVQA	+0.155	0.281	0.186	72% (vs 53%)
GQA	+0.036	0.304	0.271	58% (vs 53%)

Reading: high-edge units are demoted $\sim 12\times$ more strongly on documents than on object scenes; on DocVQA, 92% of the units post drops that pre keeps are above the per-image median, versus 35% for the reverse group. This is consistent with structures needed by text-centric recognition, but Sobel energy is neither an OCR box nor a text-region mask. The corresponding recorded token-survival examples are promoted to body Figure 3.

S3. Retention curve — full number strings ($n = 200$, both Qwen models)

pre-post Δ at 75% / 50% / 25% retention (the body § 5.4 prints endpoints only). Evidence record: audit index entry for the retention-curve ablation.

Model	Benchmark	Δ @75%	Δ @50%	Δ @25%
Qwen3-VL-8B	TextVQA	+8.7 pp	+29.0 pp	+38.3 pp
Qwen3-VL-8B	DocVQA	−1.3 pp	+5.9 pp	+26.5 pp
Qwen2.5-VL-7B	TextVQA	+7.0 pp	+15.3 pp	+32.0 pp
Qwen2.5-VL-7B	DocVQA	−2.1 pp	−1.4 pp	+18.8 pp

Reading: the two stages are indistinguishable under shallow compression (DocVQA post even leads 1–2 pp within noise at 75%); the pre-merger lead emerges and widens monotonically with depth. Same shallow regime as the Qwen2.5-VL 600k-cap reversal (body § 5.4 boundary; § 8 item 4). The Qwen3-VL curves are visualized in Supplementary Figure 5 (bottom).

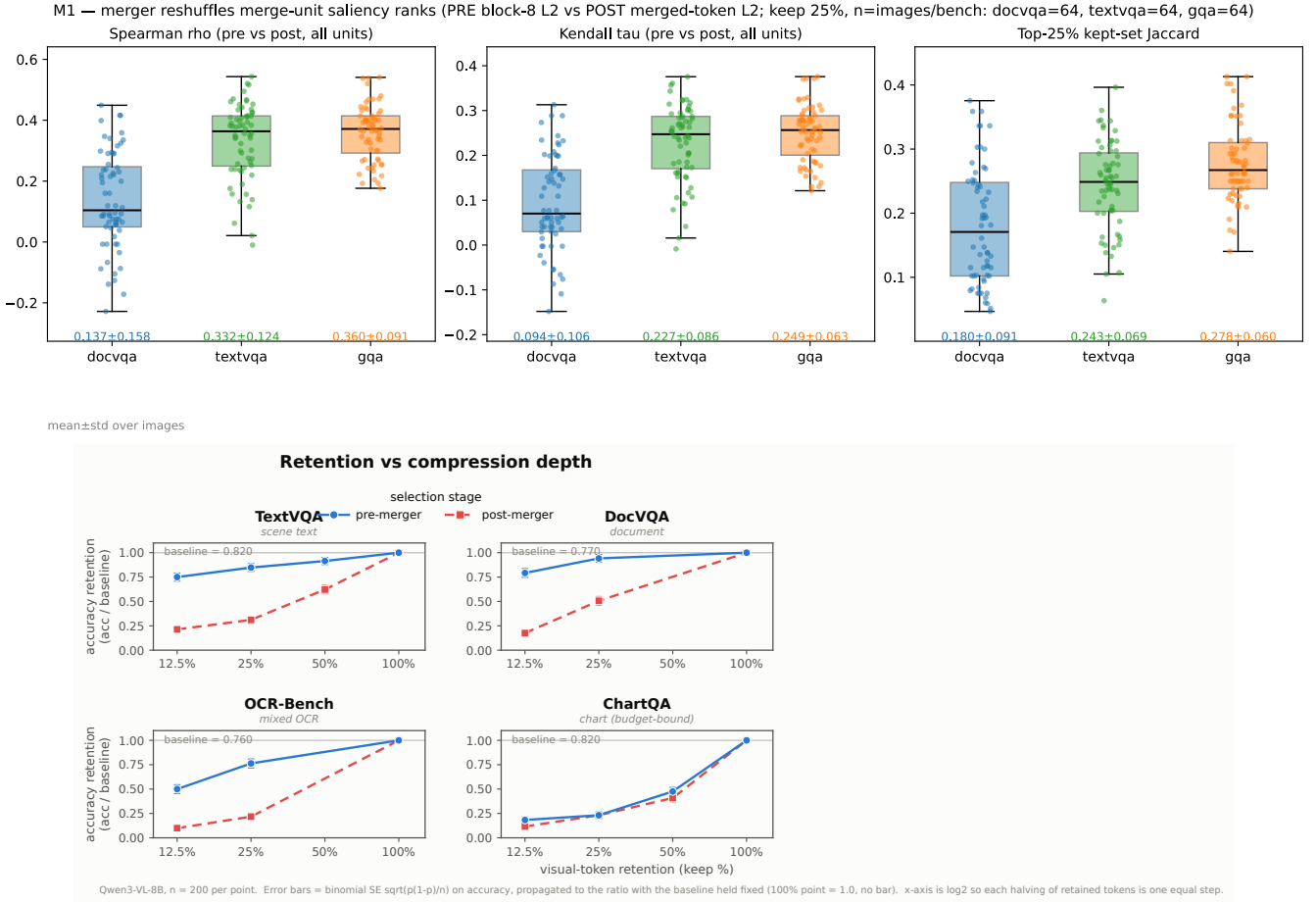


Figure 5: Quantitative mechanism and depth diagnostics. Top: per-image pre/post rank agreement on the deterministic seed-0 M1 sample ($n = 64$ per benchmark); boxes and points show the distributions, with mean±standard deviation annotated. Bottom: normalized accuracy as compression deepens on Qwen3-VL-8B ($n = 200$ per point). Pre-merger selection preserves substantially more text-dense accuracy, while ChartQA is primarily budget-limited. The stale GQA depth panel is omitted because the final campaign provides only a full-split 25% point. Error bars are descriptive rate-approximation standard errors; primary inference uses paired metric-level bootstrap and sign-flip tests.

S4. Paired McNemar cross-checks — per cell (full splits, both Qwen models)

Binary-outcome cross-check = paired McNemar z on per-sample correctness, $z = (b - c) / \sqrt{b + c}$, with b/c the pre-only / post-only correct counts; positive z = pre-merger lead, negative = post-merger lead (GQA only). Cross-check = independent-binomial z ($\text{SE} \sqrt{\text{se}_{\text{pre}}^2 + \text{se}_{\text{post}}^2}$); “agree %” = per-question agreement of the recorded correctness indicators. Evidence record: audit index entry for the paired-test table; paired by question id, restricted to ids answered in both arms; the 200/200 ground-truth self-test is retained in the audit record.

† marks operational layer-8 GQA cells favoring post. The separate full-split matched-boundary Qwen3-VL control also favors post (−5.64 pp; S7b), so GQA is a genuine workload-conditioned reversal rather than a layer-8-only artifact. The official word-normalized GQA rescore is concordant. The smallest text-dense $|z|$ is 14.9 at 25% and 14.3 at 12.5% (Qwen2.5-VL OCRBench); McNemar remains an auxiliary binary cross-check to metric-level inference.

S4b. Cascade two-stage gate — paired McNemar counts (Qwen3-VL-8B, $n = 200$, official rescore)

Cascade gate (body § 6): single budget, Stage 1 = pre-merger RBM keeping fraction X of merger-input units, Stage 2 = FastV at LLM layer 2 dropping fraction Y of the survivors; total keep = $X \cdot (1 - Y)$. Three arms (pre-alone / fastv-alone / cascade) are compared pairwise on identical images (n_{common} matched; mean post-merger token count identical across all three arms in every cell, so the comparison is iso-budget). cas_only / pre_only / fst_only = per-sample counts correct in only the named arm of the pair. Verdict = NO-GO (cascade fails the Pareto rule on all 8 budget×benchmark cells). Evidence record: audit index entry for the cascade gate and paired McNemar counts.

Reading: cascade sits strictly between the two single methods and is dominated by the better arm on every cell — pre-alone on OCR-Bench ($z = -5.15 / -5.01$ vs pre at 25% / 12.5%; the FastV stage destroys exactly the raw-patch information Stage 1 was built to protect) and FastV-alone on TextVQA / GQA ($z = -2.34 / -2.43$ vs fst at 25%). The single positive cas-vs-pre z (DocVQA 25%, +2.07) is paired with a loss to FastV (−1.50), so the “strictly beats both arms” condition holds nowhere. Two training-free selectors composed in series do not complement (body § 6).

S4c. Selector-scorer and stage-router gates (Qwen3-VL-8B, official rescore)

S4c.1 Frequency-aware selector (body § 6e): the mechanism-side probe of body § 3, $s_{\text{freq}}(u) = \alpha z(\|f_u\|_2) + \beta z(\text{Var}_u)$, tuned in two grid rounds on TextVQA₅₀₀ at 25% retention ($\alpha=1, \beta=0.6$; official VQA-acc). The variance term helps only the most text-dense cell and only under aggressive compression; everywhere else it is neutral or negative, so no no-regression-across-four-benchmarks gate passes. Evidence record: anonymized audit index entry for the frequency-aware selector.

S4c.2 Adaptive stage router (body § 6f): query-blind per-image workload detector on merger-input features — high-frequency tail ratio $r_{\text{HF}} = \Pr[\text{Var}_u > \text{Var} + \sigma_{\text{Var}}]$ and 20-bin L2-entropy $H(\|f_u\|_2)$ — routing each image to PRE (RBM, rank-before-merge) or POST (post-merger L2), both iso-budget. κ is the kept fraction; the winner of a $r_{\text{HF}} \times \tau_{\text{ent}}$ grid on TextVQA₅₀₀ ($r_{\text{HF}}=0.08, \tau_{\text{ent}}=2.0$) routes every image to PRE, reproducing plain RBM bit-for-bit: text benchmarks show 0.00 pp regression but GQA gain is also +0.00 pp (below the required +1 pp). The $r_{\text{HF}}=0.13$ threshold does route some scene images to POST but reduces TextVQA by 7 pp (51/320 flipped) while flipping only 22/243 GQA images (0.428 vs RBM 0.438). GQA and TextVQA r_{HF} distributions overlap (means 0.127 vs 0.142), so no single global threshold reconciles the two acceptance criteria. Evidence record: audit index entry for the adaptive stage-router gate and workload-signal analysis.

S4c.3 Exploratory learned boundary scorer (body § 6): we test whether a light-weight learned selector can improve on plain L2 without moving selection past the merger. A small MLP receives only pre-merger boundary features available at inference time: unit L2 norm, Sobel edge energy, local feature variance, normalized position, and neighborhood statistics. Its teacher is the post-merger importance ordering captured on separate training images; evaluation uses disjoint held-out benchmark samples and the official task scorers. Both learned and L2 arms retain exactly 25% of merger-input units and otherwise share the same Qwen3-VL-8B inference configuration.

The asymmetric task result is diagnostic rather than a method improvement. M2 independently shows that the post-merger ordering demotes edge-rich units; distillation transfers this preference into a pre-merger scorer and causes a 16.0-pp TextVQA loss. The positive GQA result shows that the learned head is capable of changing task behavior, ruling out a trivial “model did not learn” explanation. We therefore retain plain L2 and do not use the learned scorer in any headline table.

S5. Descriptive rate-scale errors + skip accounting (main full-split Qwen blocks)

For binary exact-match/containment outcomes, $\sigma = \sqrt{p(1-p)/n}$ is the Bernoulli standard error. For VQA accuracy and ANLS, which retain partial credit, applying the same expression to the printed mean is only a descriptive rate-scale approximation, not an inferential error bar. All claims use the paired metric-level bootstrap and sign-flip tests in the main paper. Evidence record: audit index entry for the Qwen full-split table and skip accounting.

- Qwen3-VL none DocVQA: re-run at max-model-len 49,152, native resolution, **0/5349 skips** (ANLS 0.956).
- OCR-Bench none cells skip **18/1000 (Qwen3-VL) / 24/1000 (Qwen2.5-VL)** long-OCR images on context overrun, scored 0 (conservative); compressed cells skip ≤ 5 . The uncompressed anchor is therefore understated and the relative gaps are conservative.
- Qwen2.5-VL OCR-Bench uses a common 4 M-pixel cap and equal per-sample token counts in both arms: mean effective prompt length 229.6 at 25% (480 vs 182, +298 points) and 131.5 at 12.5% (328 vs 70, +258 points).

Per-cell descriptive σ values (full splits, computed from the printed official metric p and full-split n):

These values provide scale and skip-accounting context only; they are not used to test partial-credit metric differences. The matched-boundary control in S7b confirms that GQA significantly favors post, unlike the three text-centric workloads. “Qwen3-VL none-DocVQA native-resolution anchor: full-split run partly skipped on context overrun (re-run at max-model-len 49,152 gives ANLS 0.956, 0/5349 skips); σ not printed for the partial native cell.” GQA evaluated at $\kappa = 0.25$ only on the full split (no full-split $\kappa = 0.125$ cells; body footnote c).

- Qwen3-VL OCR-Bench none = 760/1000 over 982 attempted (18 skips scored 0); compressed @25%: 547 (pre) / 184 (post), skip ≤ 5 — ratio 3.0×, widening to 6.6× at 12.5% (350 vs 53).

S6. $n = 200$ cross-split consistency (dev-subset vs full-split direction)

The $n = 200$ dev scope used by the regime-map and mechanism probes is direction-consistent with the full splits. Evidence record: audit index entry for the cross-split consistency campaign. $\Delta_{25} = \text{pre} - \text{post}$ at $\kappa = 0.25$.

Reading: all four text-dense cells agree in sign (pre-merger lead) across the two scopes; magnitudes are within run noise (the $n = 200$ subset has roughly 25-fold fewer samples and about fivefold larger sampling error than the full TextVQA/DocVQA splits, so it is retained only as a direction check, never as the headline). The full Qwen3-VL $n = 200$ cross-method view (RBM vs FastV vs Pyramid) and the compact Qwen2.5-VL $n = 200$ cross-generation matrix appear in the main paper’s regime table and the archived audit index, respectively.

S7. Selector invariance (second scorer family, Qwen3-VL)

The stage law under a second, non-L2 scorer (global-centroid attention). Evidence record: audit index entry for the selector-invariance campaign.

Reading: invariant in sign and magnitude on Qwen3-VL — the effect is not an L2 artifact. On Qwen2.5-VL the L2 sign is invariant but the attention proxy degrades (proxy-family specificity, not a counter-example); L2 remains the paper’s selector, frozen as plain RBM (§ 3.2).

S7b. Full-split matched-boundary control — pre-final (Qwen3-VL-8B)

The body control separates operational layer-8 RBM from the main-merger stage axis. Pre-final ranks L2 on final ViT-block features immediately before the main merger; post ranks its output. Deepstack mergers are untouched. Both arms use greedy decoding, $\kappa = 25\%$, zero skips, identical sample IDs, and equal per-sample effective prompt lengths.

Reading: the matched boundary significantly favors pre on TextVQA, DocVQA, and OCRBench, but significantly favors post on GQA. All four paired tests remain significant after Holm correction within this family. Operational layer-8 RBM improves GQA over pre-final (0.4488 vs 0.4207) but remains below post (0.4771); the stage preference is workload-conditioned.

S7c. No-merger negative control (LLaVA-1.5)

LLaVA-1.5-7B uses a per-token MLP projector rather than 2×2 spatial merging. At 25% retention ($n = 200$), pre-MLP vs post-MLP ranking shows no significant difference: TextVQA $\Delta = -2.17$ pp (none 0.452, pre 0.087, post 0.108; McNemar $p = 0.125$) and GQA $\Delta = -0.5$ pp (none 0.575, pre 0.375, post 0.380; $p = 1.0$). This single control is

Model	Benchmark	κ	Δ (official)	McNemar z (check)	b (pre-only)	c (post-only)	agree %	indep-binom z
Qwen3-VL-8B	TextVQA	.25	+38.4 pp	+43.0	2349	183	49.4	42.3
Qwen3-VL-8B	DocVQA	.25	+24.3 pp	+34.7	1601	150	67.3	27.1
Qwen3-VL-8B	OCR-Bench	.25	+363 pts	+16.7	422	56	52.2	18.2
Qwen3-VL-8B	GQA	.25	-2.8 pp $\dagger$	-8.0	834	1196	83.9	-4.5
Qwen3-VL-8B	TextVQA	.125	+34.0 pp	+40.8	2101	161	54.8	39.9
Qwen3-VL-8B	DocVQA	.125	+24.9 pp	+33.1	1437	127	70.8	32.1
Qwen3-VL-8B	OCR-Bench	.125	+297 pts	+16.0	323	25	65.2	17.8
Qwen2.5-VL-7B	TextVQA	.25	+26.1 pp	+30.8	1682	309	60.2	27.3
Qwen2.5-VL-7B	DocVQA	.25	+11.0 pp	+15.6	1406	693	60.8	11.6
Qwen2.5-VL-7B	OCR-Bench	.25	+298 pts	+14.9	349	51	60.0	14.9
Qwen2.5-VL-7B	GQA	.25	-2.6 pp $\dagger$	-5.7	892	1150	83.8	-4.1
Qwen2.5-VL-7B	TextVQA	.125	+27.8 pp	+33.1	1836	305	57.2	29.1
Qwen2.5-VL-7B	DocVQA	.125	+21.0 pp	+25.6	1325	294	69.7	23.3
Qwen2.5-VL-7B	OCR-Bench	.125	+258 pts	+14.3	291	33	67.6	15.3

total keep κ	bench	n	pre	fst	cas	z (cas vs pre)	cas_only	pre_only	z (cas vs fst)	cas_only	fst_only
25%	TextVQA	200	0.5967	0.6800	0.5950	-0.152	21	22	-2.335	18	35
25%	DocVQA	200	0.4239	0.5183	0.4779	+2.066	38	22	-1.500	26	38
25%	OCR-Bench	181	0.5801	0.1713	0.3978	-5.154	4	37	+6.112	43	2
25%	GQA	200	0.4150	0.4900	0.4050	-0.333	17	19	-2.429	16	33
12.5%	TextVQA	200	0.5000	0.4450	0.4050	-2.540	21	41	-0.896	27	34
12.5%	DocVQA	200	0.2980	0.2779	0.2070	-2.012	31	49	-2.064	16	30
12.5%	OCR-Bench	181	0.3591	0.0884	0.1713	-5.013	6	40	+3.441	17	2
12.5%	GQA	200	0.3950	0.4550	0.3500	-1.286	20	29	-3.130	12	33

bench@ κ	freq($\alpha=1, \beta=0.6$)	RBM (L2)	Post-L2	Δ freq-RBM
TextVQA @25%	0.6173	0.6053	0.2073	+1.20
TextVQA @50%	0.6780	0.6733	0.3873	+0.47
DocVQA @25%	0.4903	0.5063	0.2317	-1.60
DocVQA @50%	0.6271	0.6212	0.5487	+0.59
OCR-Bench @25%	0.7160	0.7380	0.1700	-2.20
OCR-Bench @50%	0.8220	0.8220	0.4980	0.00
GQA @25%	0.4220	0.4380	0.4540	-1.60
GQA @50%	0.4860	0.4840	0.5000	+0.20

Table 4: Frequency-aware selector (official metrics, $n = 500$; TextVQA=1.2 pp gain at $\kappa=25\%$, baseline-equal elsewhere, strong regression on the other three benchmarks at 25%).

Benchmark	Learned POST teacher	Plain L2	Δ learned-L2
TextVQA	0.435	0.595	-16.0 pp
GQA	0.550	0.515	+3.5 pp

Table 5: Held-out learned-scorer evaluation at 25% retention ($n = 200$ per benchmark; official metrics). The scorer can imitate a post-merger teacher, but the resulting ordering is strongly anti-task on TextVQA and therefore fails the cross-workload no-regression gate.

consistent with native spatial merging being relevant to the large text-centric gaps, but does not establish necessity.

S8. Merger tap points and M-RoPE configuration details (supports §3)

- **Qwen3-VL-8B**: the first deepstack merger has index 0 and the visual indexes are [8, 16, 24], so unit scores use **ViT block-8 outputs** (intermediate depth); the single resulting mask is shared by all mergers, including the main

merger that consumes the final-block output. The tap-point and call-order verification is retained in the audit index.

- **Qwen2.5-VL-7B**: no deepstack; scores are computed on the **main merger input (final ViT block output)**. `mask_computed_at='main'`.
- **InternVL3-8B**: the pixel-shuffle downsampling stage is the merger; L2 taps its input.
- **M-RoPE fix**: Qwen2.5-VL uses a block M-RoPE layout and Qwen3-VL an interleaved one; under pruning the position cursor must advance by the actual surviving count. A family-scoped fix (bit-degrading to the original at full retention) makes the Qwen2.5-VL compression path well-formed; the block-vs-interleaved contrast diagnoses why naive pruning degrades one generation more strongly than the other.
- **r2c harness**: HF transformers eager attention, `--mrope` (family-agnostic correct positions) on all same-scope cells; the Qwen2.5-VL $\times$ pre vllm-mimic positional degeneration (repeated “addCriterion...” outputs $\rightarrow$ pseudo-zero scores) was detected and replaced before any reported number; on Qwen3-VL native vs mimic give 0.575 vs 0.525 (same ordering). The harness and position-layout verification are retained in the audit index.

S9. Per-cell reproducibility index

The filenames below define the anonymous artifact manifest and the audit records required for submission. The archive must be checked against this index before external upload; the tables remain self-contained summaries of the reported cells.

S9.1 InternVL3-8B full-split cells (16/16)

Model OpenGVLab/InternVL3-8B, vLLM 0.19, bf16, eager attention, greedy; $r = \text{TOTAL drop}$ (0.750 $\rightarrow \kappa = 25\%$, 0.875 $\rightarrow \kappa = 12.5\%$); skip = context-overflow cells scored 0. Evidence record: audit index entry for the InternVL3 full-split metrics and per-cell skip accounting.

(Per-cell artifact identifiers are specified by the anonymous manifest. Pre-merger leads on every text-dense benchmark at both budgets — e.g. DocVQA ANLS 0.7284 vs 0.382 at $\kappa = 25\%$; GQA is the near-null object-centric case, 0.5993 vs 0.6031 — replicating the stage law on the pixel-shuffle merger design.)

S9.2 FastV-k3 same-scope cells (8/8)

FastV one-shot attention pruning at LLM layer $K = 3$, iso-budget $r = 0.75$ ($\kappa = 25\%$); HF-transformers 4.57.6 eager harness, official metrics. Official values are rescored outputs in the audit index; n , effective prompt length, skip, engine, and per-cell identifiers are audited there. The harness’s raw acc field is not used as the official score.

Model	Benchmark	n	none	pre@25%	post@25%	pre@12.5%	post@12.5%
Qwen3-VL-8B	TextVQA	5000	0.0051	0.0069	0.0059	0.0071	0.0048
Qwen3-VL-8B	DocVQA	5349	— ^a	0.0068	0.0058	0.0065	0.0042
Qwen3-VL-8B	OCR-Bench	1000	0.0135	0.0157	0.0123	0.0151	0.0071
Qwen3-VL-8B	GQA	12578	0.0043	0.0044	0.0045	— ^c	— ^c
Qwen2.5-VL-7B	TextVQA	5000	0.0049	0.0065	0.0070	0.0069	0.0066
Qwen2.5-VL-7B	DocVQA	5349	0.0030	0.0066	0.0068	0.0068	0.0059
Qwen2.5-VL-7B	OCR-Bench	1000	0.0122	0.0158	0.0122	0.0148	0.0081
Qwen2.5-VL-7B	GQA	12578	0.0044	0.0044	0.0044	— ^c	— ^c

Model	Bench	full Δ_{25}	$n = 200$ Δ_{25}	sign
Qwen3-VL	TextVQA	+38.4 pp	+38.3 pp	✓ pre
Qwen3-VL	DocVQA	+24.3 pp	+26.5 pp	✓ pre
Qwen2.5-VL	TextVQA	+26.1 pp	+32.0 pp	✓ pre
Qwen2.5-VL	DocVQA	+11.0 pp	+18.8 pp	✓ pre

Selector	Benchmark	pre	post	Δ
L2	TextVQA	0.598–0.605	0.215–0.222	$\approx +38$ pp
GC-attn	TextVQA	0.553	0.200	+35.3 pp
GC-attn	TextVQA/DocVQA (2n)	—	—	+35.5 / +24.4 pp

(The two OCRBench rows are the final full-1000, 4M-cap controls. Their 79/88 skips are symmetric with RBM, leaving 921/912 common answered samples for paired inference. Non-OCR rows retain their original printed scopes.)

S9.3 RBM same-scope cells (5/5)

RBM (pre-merger L2) at iso-budget $r = 0.75$, HF-transformers eager, `--mropenative` (family-agnostic correct positions), matched to the FastV-k3 cells above. Official values, effective prompt lengths, skips, and per-cell identifiers are recorded in the audit index.

S9.4 Cascade gate cells (24/24, Qwen3-VL-8B)

Three arms $\times$ two budgets $\times$ four benchmarks; $n = n_{\text{common}}$ (matched by question id); official = `acc_official` from the paired rescoring; `ptid` = mean post-merger token count (identical across the three arms in every cell). Evidence record: audit index entry for the cascade paired summary and per-arm cells. (McNemar counts for these cells: S4b.)

(Per-cell artifact identifiers are specified by the anonymous manifest. OCR-Bench $n_{\text{common}} = 181$ after 19 long-OCR skips; all other cells $n = 200$. All 24 expected cells are present.)

S10. Qualitative same-image comparison — measured pre- vs post-merger L2 selection

For four representative inputs we render, on the *same* processor-resized image, unit grid, 25% budget and final visual-token count, the original image with its full merge-unit grid, the RBM (pre-merger L2 top- k) kept mask, the post-merger L2 top- k kept mask (same L2 selector, same budget), and a four-state difference map (kept-by-both / RBM-only / post-only / dropped-by-both), encoded by colour *and* hatch/outline so the states remain distinguishable in greyscale). Each panel reports the measured top- k Jaccard index $J = |\text{pre} \cap \text{post}| / |\text{pre} \cup \text{post}|$ and the full-ranking Spearman ρ between the pre- and post-merger L2 orderings (computed from the stored NPZ rank arrays; values cross-checked against the validation report, $|\Delta| < 10^{-3}$). All scores are the target model’s own merger-input / merger-output feature L2 norms (Qwen3-VL-8B-Instruct, `enforce_eager`, `max_pixels = 1.5M`); no CLIP, SigLIP, attention map, or fabricated saliency is used. The FastV layer-2 panel is intentionally absent (no user-supplied question exists for these images; a query-conditioned mask cannot be measured without one) and is recorded as `fastv_status: skipped_missing_question`.

The consistently low Jaccard (0.25–0.35) and moderate Spearman (0.30–0.48) across the four shown inputs are the image-level signature of the body paper’s mechanism result: the learned 2×2 merger *reshuffles* the unit ranking, so selecting before versus after it on the same image with the same selector and budget keeps largely *different* units. The four-state maps show that the disagreement is spatially structured (contiguous RBM-only and post-only regions), not uniform noise, and that the “kept-by-both” intersection is a sparse minority of the budget.

Packaging note: this source package contains the rendered summaries above, not the run-level anonymous artifact indexed in S9. That archive must be restored, sanitized, checksummed, and verified before submission.

Benchmark (metric)	none	pre-final	post	Δ [95% CI]	<i>n</i>
TextVQA (VQA-acc)	0.8443	0.4985	0.2217	+27.68 pp [+26.21, +29.19]	5000
DocVQA (ANLS)	0.9562	0.2836	0.2377	+4.59 pp [+3.43, +5.78]	5349
OCRBench (/1000)	760	419	184	+235 pt [+199, +271]	1000
GQA (exact match)	0.6165	0.4207	0.4771	-5.64 pp [-6.42, -4.86]	12578

Body location	Cells	Artifact record	Audit summary
Main full-split Qwen blocks	26/26	audit: main-qwen	full-split metrics and skip accounting
Main full-split InternVL3 block	16/16	audit: main-internvl3	full-split metrics and skip accounting
Regime-map FastV-k3	8	audit: regime-fastv-k3	same-scope rescored cells
Regime-map RBM same-scope	5	audit: regime-rbm	same-scope rescored cells
Cascade gate	24/24	audit: cascade	paired gate statistics
Efficiency study	7 configs	audit: efficiency	throughput and latency records
§ 5 kept-set Jaccard / swap	8	audit: mechanism-swap	kept-set and ordering audit
§ 5.1-5.2 M1/M2	$n = 64 \times 3$	audit: mechanism-m1-m2	rank and edge-energy measurements
§ 5.4 retention curve	$n = 200 \times 4$	audit: retention-curve	depth ablation records
§ 6 (a)-(f) prespecified extensions	—	audit: negative-results	gates and extension records
Learned-scorer probe	$n = 200 \times 2$	audit: learned-scorer	model, held-out splits, predictions, and official rescoring
§ 8 600k reversal	3	audit: low-cap-reversal	configuration-boundary records
Engine fairness	$r=0$ 8/8; none 16/16	audit: engine-parity	zero-drop and none-cell checks
PyramidDrop iso-25% degradation	$n = 500$	audit: pyramiddrop-iso25	iso-budget baseline record

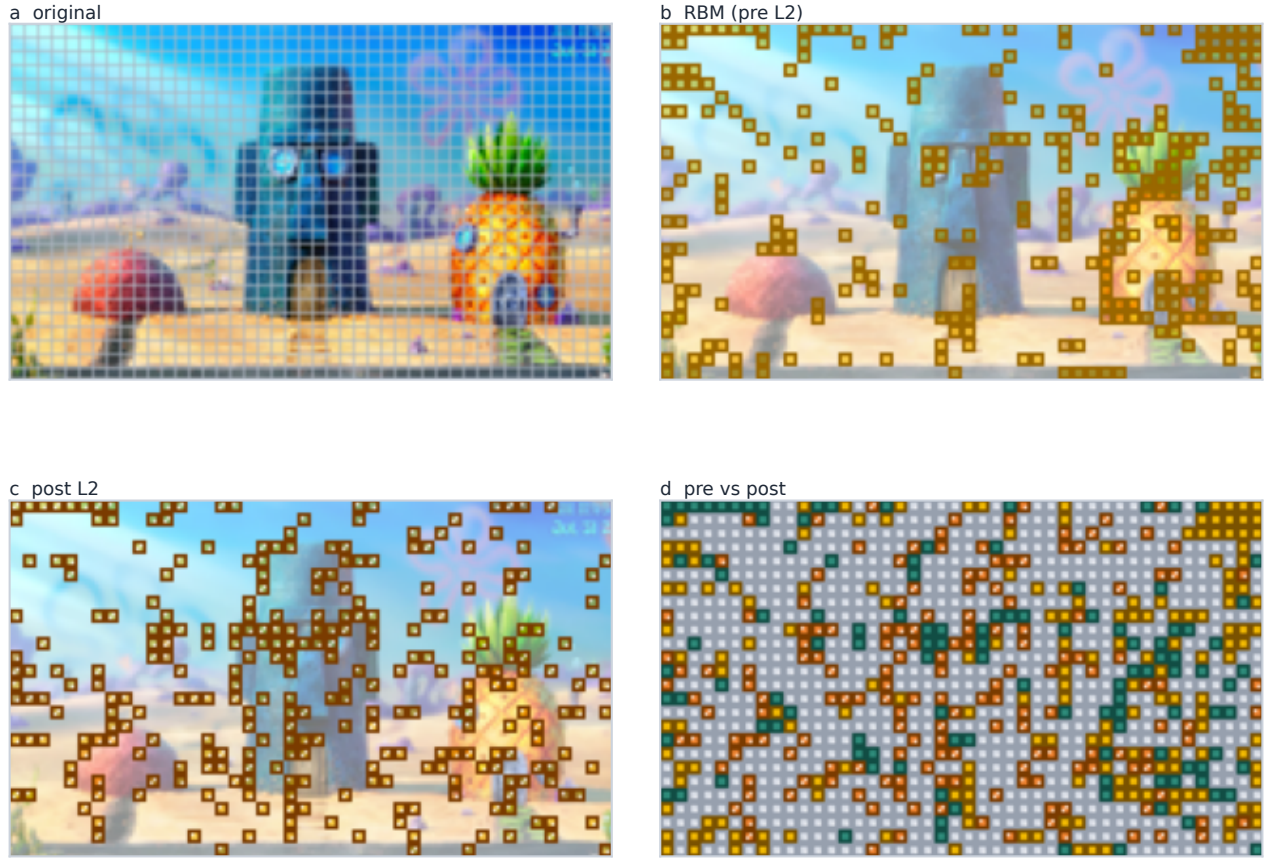
cell	mode	<i>r</i>	bench	<i>n</i>	official	metric	ptid	skip	JSON
none_docvqa	none	0.000	docvqa	5349	0.9221	ANLS	3230.8	0	internvl3_none_docvqa_r0.000_full.json
none_gqa	none	0.000	gqa	12578	0.6293	acc-official	2557.2	0	internvl3_none_gqa_r0.000_full.json
none_ocrbench	none	0.000	ocrbench	1000	0.852	Final/1000=852	2040.1	0	internvl3_none_ocrbench_r0.000_full.json
none_textvqa	none	0.000	textvqa	5000	0.8338	VQA-acc	2562.9	0	internvl3_none_textvqa_r0.000_full.json
pre_docvqa_r0.750	pre	0.750	docvqa	5349	0.7284	ANLS	859.5	0	internvl3_pre_docvqa_r0.750_full.json
pre_docvqa_r0.875	pre	0.875	docvqa	5349	0.5054	ANLS	464.3	0	internvl3_pre_docvqa_r0.875_full.json
pre_gqa_r0.750	pre	0.750	gqa	12578	0.5993	acc-official	690.1	0	internvl3_pre_gqa_r0.750_full.json
pre_ocrbench_r0.750	pre	0.750	ocrbench	1000	0.753	Final/1000=753	555.4	0	internvl3_pre_ocrbench_r0.750_full.json
pre_textvqa_r0.750	pre	0.750	textvqa	5000	0.789	VQA-acc	690.5	0	internvl3_pre_textvqa_r0.750_full.json
pre_textvqa_r0.875	pre	0.875	textvqa	5000	0.723	VQA-acc	378.4	0	internvl3_pre_textvqa_r0.875_full.json
post_docvqa_r0.750	post	0.750	docvqa	5349	0.382	ANLS	859.5	0	internvl3_post_docvqa_r0.750_full.json
post_docvqa_r0.875	post	0.875	docvqa	5349	0.2451	ANLS	464.3	0	internvl3_post_docvqa_r0.875_full.json
post_gqa_r0.750	post	0.750	gqa	12578	0.6031	acc-official	690.1	0	internvl3_post_gqa_r0.750_full.json
post_ocrbench_r0.750	post	0.750	ocrbench	1000	0.321	Final/1000=321	555.4	0	internvl3_post_ocrbench_r0.750_full.json
post_textvqa_r0.750	post	0.750	textvqa	5000	0.4148	VQA-acc	690.5	0	internvl3_post_textvqa_r0.750_full.json
post_textvqa_r0.875	post	0.875	textvqa	5000	0.3064	VQA-acc	378.4	0	internvl3_post_textvqa_r0.875_full.json

cell / artifact id	model	bench	<i>n</i>	official	prompt len.	skip	engine
r2b_qwen3vl_fastv_k3_textvqa_r0.75_full5000	Qwen3-VL	textvqa	5000	0.7771	213.7	5	hf eager
r2b_qwen3vl_fastv_k3_gqa_r0.75_full12578	Qwen3-VL	gqa	12578	0.5376	96.8	0	hf eager
r2b_qwen3vl_fastv_k3_docvqa_r0.75_n500	Qwen3-VL	docvqa	200	0.5863	176.0	0	hf eager
p1_qwen3vl_fastv_ocrbench_k3_full	Qwen3-VL	ocrbench	1000	0.413	110.7	79	hf eager
r2b_qwen2vl_fastv_k3_textvqa_r0.75_n200	Qwen2.5-VL	textvqa	200	0.7467	282.1	0	hf eager
r2b_qwen2vl_fastv_k3_gqa_r0.75_n200	Qwen2.5-VL	gqa	200	0.475	126.8	0	hf eager
r2b_qwen2vl_fastv_k3_docvqa_r0.75_n200	Qwen2.5-VL	docvqa	200	0.4852	231.4	0	hf eager
p1_qwen2vl_fastv_ocrbench_k3_full	Qwen2.5-VL	ocrbench	1000	0.295	134.7	88	hf eager

cell / artifact id	model	bench	<i>n</i>	official	prompt len.	skip	engine
r2c_qwen2vl_pre_r0.75_textvqa_n200	Qwen2.5-VL	textvqa	200	0.6683	282.1	0	hf eager
r2c_qwen2vl_pre_r0.75_gqa_n200	Qwen2.5-VL	gqa	200	0.520	126.8	0	hf eager
r2c_qwen2vl_pre_r0.75_docvqa_n200	Qwen2.5-VL	docvqa	200	0.5062	231.4	0	hf eager
p1_qwen2vl_pre_ocrbench_r25_full	Qwen2.5-VL	ocrbench	1000	0.382	134.7	88	hf eager
p1_qwen3vl_pre_ocrbench_r25_full	Qwen3-VL	ocrbench	1000	0.559	110.7	79	hf eager

arm	κ	bench	n	official	ptid	JSON
pre-alone (RBM)	25%	textvqa	200	0.5967	212.8	gate_pre25_textvqa.json
fastv-alone	25%	textvqa	200	0.6800	212.8	gate_fst25_textvqa.json
cascade	25%	textvqa	200	0.5950	212.8	gate_cas25_textvqa.json
pre-alone (RBM)	25%	docvqa	200	0.4239	176.0	gate_pre25_docvqa.json
fastv-alone	25%	docvqa	200	0.5183	176.0	gate_fst25_docvqa.json
cascade	25%	docvqa	200	0.4779	176.0	gate_cas25_docvqa.json
pre-alone (RBM)	25%	ocrbench	181	0.5801	114.8	gate_pre25_ocrbench.json
fastv-alone	25%	ocrbench	181	0.1713	114.8	gate_fst25_ocrbench.json
cascade	25%	ocrbench	181	0.3978	114.8	gate_cas25_ocrbench.json
pre-alone (RBM)	25%	gqa	200	0.4150	95.4	gate_pre25_gqa.json
fastv-alone	25%	gqa	200	0.4900	95.4	gate_fst25_gqa.json
cascade	25%	gqa	200	0.4050	95.4	gate_cas25_gqa.json
pre-alone (RBM)	12.5%	textvqa	200	0.5000	120.7	gate_pre12_textvqa.json
fastv-alone	12.5%	textvqa	200	0.4450	120.7	gate_fst12_textvqa.json
cascade	12.5%	textvqa	200	0.4050	120.7	gate_cas12_textvqa.json
pre-alone (RBM)	12.5%	docvqa	200	0.2980	103.7	gate_pre12_docvqa.json
fastv-alone	12.5%	docvqa	200	0.2779	103.7	gate_fst12_docvqa.json
cascade	12.5%	docvqa	200	0.2070	103.7	gate_cas12_docvqa.json
pre-alone (RBM)	12.5%	ocrbench	181	0.3591	68.7	gate_pre12_ocrbench.json
fastv-alone	12.5%	ocrbench	181	0.0884	68.7	gate_fst12_ocrbench.json
cascade	12.5%	ocrbench	181	0.1713	68.7	gate_cas12_ocrbench.json
pre-alone (RBM)	12.5%	gqa	200	0.3950	62.3	gate_pre12_gqa.json
fastv-alone	12.5%	gqa	200	0.4550	62.3	gate_fst12_gqa.json
cascade	12.5%	gqa	200	0.3500	62.3	gate_cas12_gqa.json

Same-image iso-budget comparison - df7282e1 (Qwen3-VL-8B measured merger L2)



$N=1144$ $k=286$ (25% kept) top-k Jaccard=0.2599 Spearman $\rho(\text{pre}, \text{post ranks})=0.3141$ unit grid $26 \times 44 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2

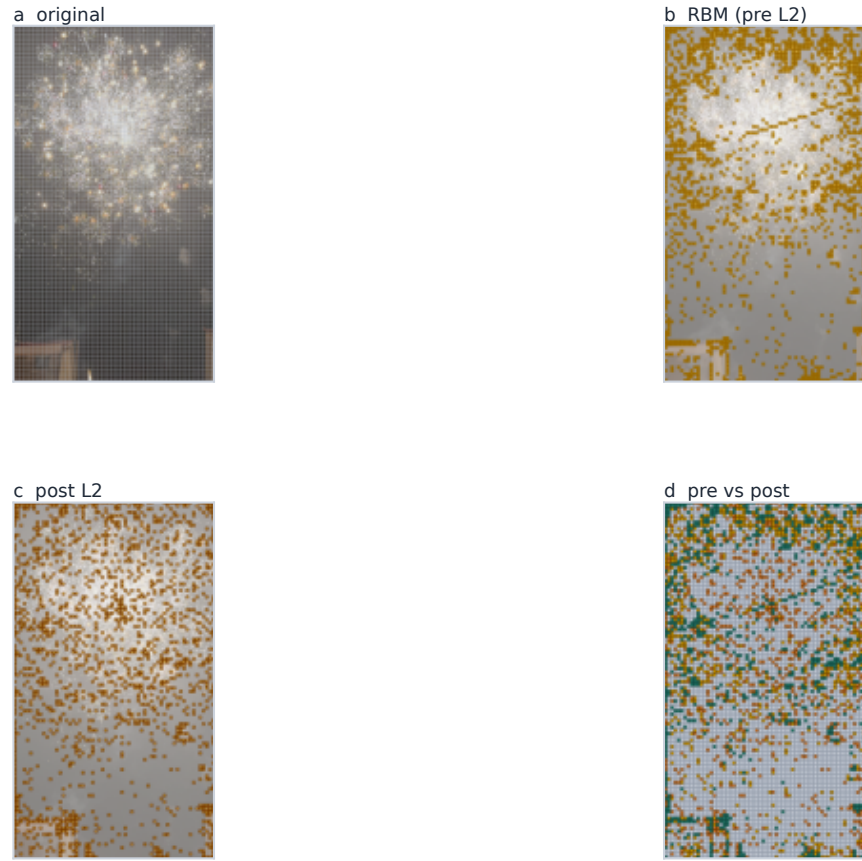
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).
 (d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

Figure 6: Sample df7282e1 ($N = 1144$, $k = 286$, $J = 0.260$, $\rho = 0.314$). Sparse scene; the four-state map shows RBM-only (amber) concentrated on object boundaries while post-only (hatched) scatters over flat regions the merger smoothed.

Same-image iso-budget comparison - 69 (Qwen3-VL-8B measured merger L2)



$N=6820$ $k=1705$ (25% kept) top-k Jaccard=0.3478 Spearman $\rho(\text{pre}, \text{post ranks})=0.4774$ unit grid $110 \times 62 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2

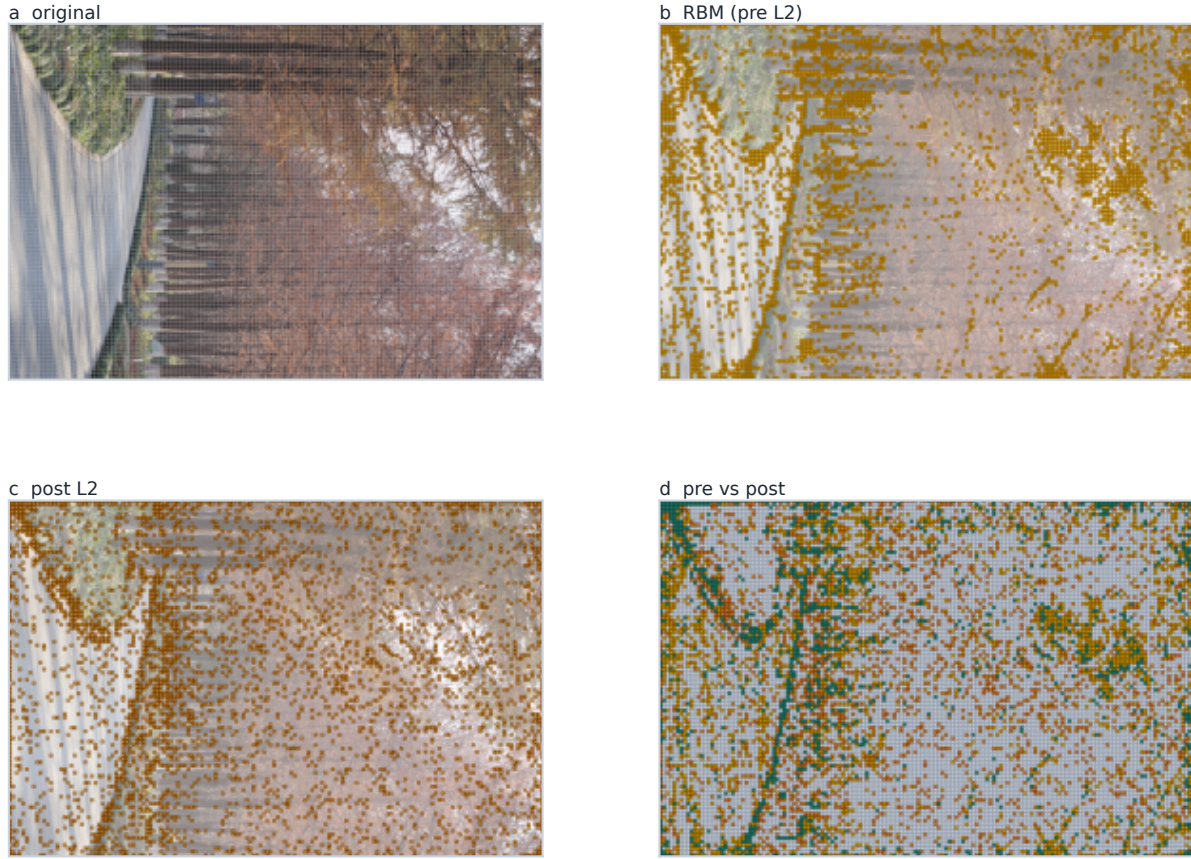
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).
(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

Figure 7: Sample **img69 ($N=6820$, $k=1705$, $J=0.348$, $\rho=0.477$). Highest overlap of the four; the merger preserves more of the pre-ranking here, yet two-thirds of the kept set still differs.**

Same-image iso-budget comparison - 72 (Qwen3-VL-8B measured merger L2)



$N=16224$ $k=4056$ (25% kept) $\text{top-k Jaccard}=0.2482$ $\text{Spearman } \rho(\text{pre}, \text{post ranks})=0.2999$ $\text{unit grid } 104 \times 156 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2

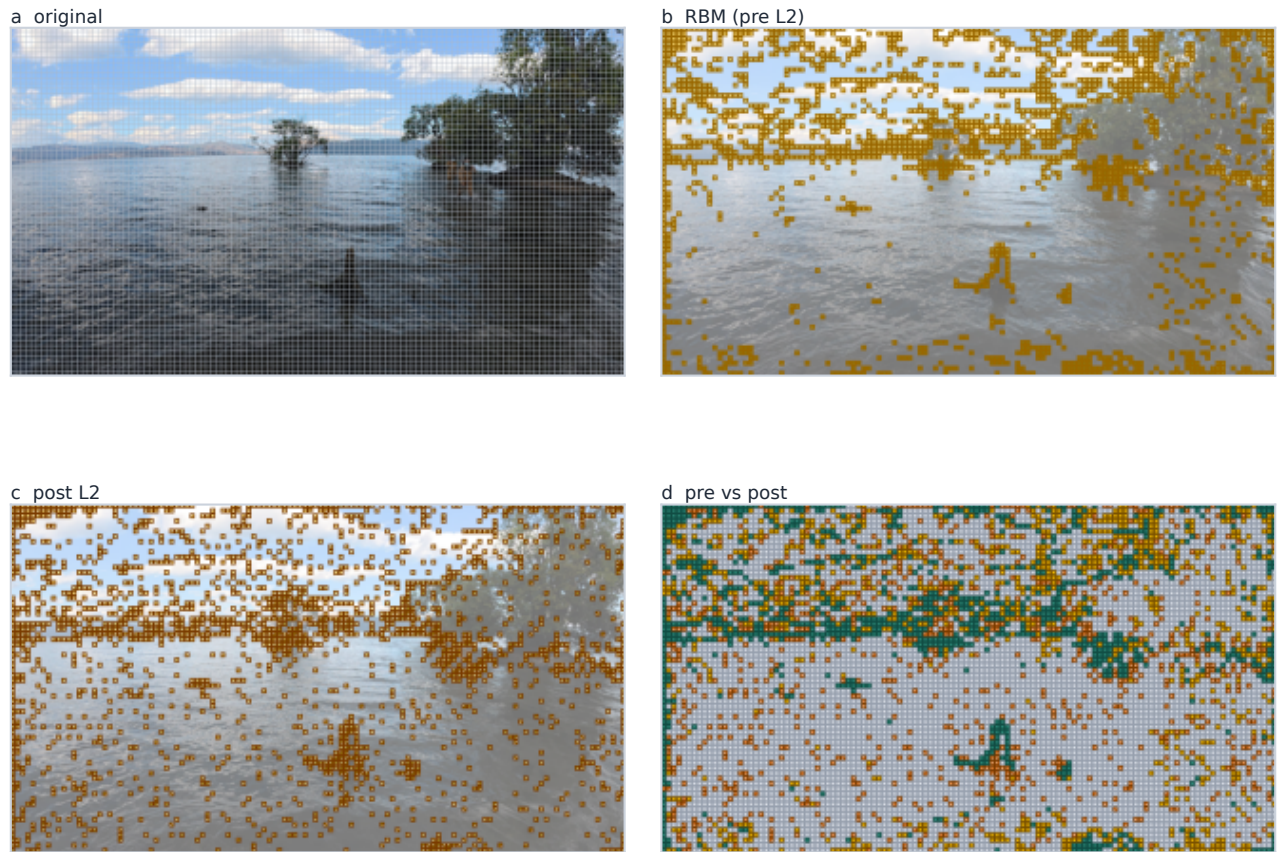
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).
(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

Figure 8: Sample img72 ($N=16224$, $k=4056$, $J=0.248$, $\rho=0.300$). Textured outdoor scene with a wall edge; the lowest overlap – the merger’s lossy aggregation most strongly reorders the high-frequency edge vs. texture units.

Same-image iso-budget comparison - 74 (Qwen3-VL-8B measured merger L2)



$N=8160$ $k=2040$ (25% kept) top-k Jaccard=0.3377 Spearman $\rho(\text{pre}, \text{post ranks})=0.3963$ unit grid $68 \times 120 = \text{grid_thw}/2$ Qwen3-VL-8B measured merger L2

FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).

(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

Figure 9: Sample `img74` ($N=8160$, $k=2040$, $J=0.338$, $\rho=0.396$). Waterscape with horizon; RBM-only units track the water surface detail that the post-merger L2 demotes in favour of merged sky blocks.